



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING
UTM Johor Bahru

Semester II 2023/2024

Subject : **SECD2613 System Analysis and Design**
Task : **Assignment 1**
Title : **Group Assignment 1 (Project Planning 5%)**
Due : **19/4/2024 5:00 PM**
Submission : **Submit softcopy (in .pdf format) via e-learning**

INSTRUCTIONS:

1. 3 ~ 4 students per group
2. All answers should be printed on and uploaded via link specified in e-learning.
3. Any plagiarism – copying directly from other groups will result in both groups will penalized by getting an 'F' grade.

Question 1 (10 Marks)

“ We could really make some changes. Shake up some people. Let them know we’re with it. Technologically, I mean,” said Ahmad, vice president for AllFoods, a wholesale dairy products distributor. “That old system should be overhauled. I think we should just tell the staff that it’s time to change.”

“Yes, but what would we actually be improving?” Abu, assistant to the vice president, asks. “I mean, there aren’t any substantial problems with the system input or output that I can see.”

Ahmad snaps, “Abu, you’re purposely not seeing my point. People out there see us as a stodgy firm. A new computer system could help change that. Change the look of our invoices. Send a report to the food store owners. Get some people excited about us as leaders in wholesale food distributing and computers.”

“Well, from what I’ve seen over the years,” Abu replies evenly, “a new system is very disruptive, even when the business really needs it. People dislike change, and if the system is performing the way it should, maybe there are other things we could do to update our image that wouldn’t drive everyone nuts in the process. Besides, you’re talking big bucks for a new gimmick.”

Ahmad says, “I don’t think just tossing it around here between the two of us is going to solve anything. Check on it and get back to me. Wouldn’t it be wonderful?”

A week later Abu enters Ahmad’s office with several pages of interview notes in hand. “I’ve talked with most of the people who have extensive contact with the system. They’re happy, Ahmad. And they’re not just talking through their hats. They know what they’re doing.”

“I’m sure the managers would like to have a newer system than the guys at Quality Foods,” Ahmad replies. “Did you talk to them?” Abu says, “Yes. They’re satisfied.”

“And how about the people in systems? Did they say the technology to update our system is out there?” Ahmad inquires insistently. “Yes. It can be done. That doesn’t mean it should be,” Abu says firmly. As the systems analyst for AllFoods,

- a. how would you assess the feasibility of the systems project Ahmad is proposing?
(2 Marks)
- b. Based on what Abu has said about the managers, users, and systems people, what seems to be the operational feasibility of the proposed project?
(2 Marks)
- c. What about the economic feasibility?
(2 Marks)
- d. What about the technological feasibility?
(2 Marks)
- e. Based on what Ahmad and Abu have discussed, would you recommend that a full-blown systems study be done? Discuss your answer in a paragraph.
(2 marks)

Question 2 (10 Marks)

Think of all the tasks that you perform when you want to propose a new computer system for student registration. Include any research, decisions, or financial issues that related to the development of new computer system. Create a Work base Schedule (WBS) that shows all the tasks, their estimated duration, and any predecessor tasks.

Question 3 (10 Marks)

Many of today's projects involve team members scattered across different time zones and in different physical locations. Moreover, the projects may have adopted an agile methodology, which reduces cycle time dramatically. Write a brief report that summarizes some of the key differences a manager would face managing this type of project, as opposed to a traditional project.

Question 4 (10 Marks)

SkorBistari leading preschool was established with the dream of bringing quality of preschools to greater high based on Islamic principles, English emphasis, and cheerful well-designed facilities. The company grows to over 53 centers all over the country. Recently, the top management has decided to consider an information system to help their growing business. You have been hired as the system analyst to handle the project. Before continue with the development of the project, you need to assess the economic feasibility from the budget that the company plans to invest. Table 1 shows the information given by the company.

Table 1: Estimated cost and expected benefits for SkorBistari

Development Costs

Hardware RM 50,000

Software RM 15,000

Training RM 30,000

Consulting RM 80,000

Data Conversion RM 40,000

Production Costs

Supplies RM 15,000 per year

IS Salaries RM 75,000 in year 1; 10% annual increase

Upgrades RM 10,000 per year over 3 years

Benefits

Improve Customer Service RM 150,000 in year 1; 20% increase annually

Increase productivity RM 150,000 per year; 25% increase annually

Assumptions Discount rate 35 %

(a) Calculate the cost-benefit estimation using the Present Value (PV) analysis (for 3 years) to assess the economic feasibility for the SkorBestari project proposal in the worksheet.
(9 marks)

(b) What is the profitability index (PI) value for this PV analysis? What is your recommendation based on the PI value and justify it.
(1 marks)

QUESTION 5 (10 MARKS)

A project has been defined to contain the following list of activities along with their required times of completion.

Activity No	Immediate Activity	Time (weeks)	Predecessors
1	Collect requirements	3	
2	Analyze processes	2	1
3	Analyze data	2	2
4	Design Process	6	2
5	Design data	3	3
6	Design Screens	2	3, 4
7	Design reports	4	4,5
8	Program	5	6, 7
9	Test and document	7	7
10	Install	2	8, 9

- a. Draw a PERT Diagram for the activities. (3 Marks)
- b. Lists all the paths. (2 Marks)
- c. Calculate and identify the critical path. (2 Marks)
- d. Construct a Gantt chart for the project defined above. (3 Marks)